Graph-Based Twitter Bot Detection and Semantic Polarization Analysis: A Dual-Module Study

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Abstract

Social media platforms are increasingly affected by automated accounts and ideological polarization. This study presents a dual-module framework in which bot detection and polarization analysis are conducted independently. The bot detection module employs a Graph Neural Network (GraphSAGE) using user metadata, behavioral attributes, and network structure. The polarization module utilizes transformer-based sentence embeddings followed by unsupervised clustering (HDBSCAN and KMeans) and dimensionality reduction using UMAP. A polarization index based on inter-cluster and intra-cluster distances is defined, along with an echo chamber strength metric. An intervention framework is introduced to evaluate the impact of random and structured perturbations on polarization. The results demonstrate strong classification performance for bot detection and clear ideological clustering in the semantic space. Intervention analysis indicates that dominant clusters significantly influence polarization levels.

Introduction:

Social media platforms play a central role in information dissemination but are increasingly vulnerable to manipulation through automated accounts and ideological polarization. Automated accounts contribute to the spread of misinformation, while polarized communities reinforce homogeneous viewpoints through echo chambers. Existing research typically addresses bot detection and polarization independently. This study presents two complementary analytical modules that examine these phenomena in parallel, enabling a structured understanding of both behavioral and semantic patterns in social media data.

Literature Review:

Bot Detection Module:

Social bot detection has been extensively studied using feature-based, text-based, and graph-based approaches. Early methods relied on handcrafted features such as user metadata, activity patterns, and temporal behavior, combined with traditional machine learning algorithms. While these approaches can achieve reasonable performance, their effectiveness is highly dependent on feature engineering and often lacks generalization across datasets.

Text-based approaches utilize natural language processing techniques to analyze tweet content. However, modern bots increasingly mimic human language behavior, reducing the effectiveness of purely content-based models.

Recent advances focus on graph-based methods, where users are represented as nodes and interactions as edges. Graph Neural Networks (GNNs), including Graph Convolutional Networks and GraphSAGE, have demonstrated state-of-the-art performance by capturing relational dependencies among users. The introduction of large-scale benchmarks such as TwiBot-22 has further accelerated research in this domain by providing richer graph structures and improved annotation quality .

More recent work explores hybrid architectures combining graph structure with semantic information. For instance, language model–GNN fusion approaches improve detection performance, particularly for sparsely connected nodes, by integrating textual and structural representations . Similarly, ensemble-based GNN methods enhance robustness by combining multiple graph representations .

Despite these advancements, many graph-based methods require complex heterogeneous graphs or multimodal inputs, increasing computational complexity.

Polarization Module:

Polarization and echo chambers in social media have been widely studied in computational social science. Prior work demonstrates that users tend to form ideologically homogeneous groups, leading to reduced exposure to diverse viewpoints and reinforcement of existing beliefs.

Most existing studies analyze polarization using network-based approaches, focusing on interaction graphs to identify community structures. However, these approaches often lack semantic understanding of content and may fail to capture nuanced ideological differences.

Recent advances leverage text embeddings and clustering techniques to analyze polarization in semantic space. Transformer-based models provide rich contextual representations of text, enabling more accurate identification of ideological clusters. However, these methods often rely on qualitative analysis or visualization and lack formal quantitative metrics for polarization measurement.

Furthermore, existing studies rarely incorporate intervention-based analysis to evaluate how polarization changes under different conditions.

Aspect	Existing Work	Proposed Work
Bot Detection	Feature / text / graph models	GraphSAGE with feature + structure
Graph Complexity	Often heterogeneous	Simplified graph
Polarization Analysis	Mostly qualitative / network-based	Quantitative + semantic clustering
Metrics	Limited	Defined polarization + echo metrics
Intervention Analysis	Rare	Explicitly modeled

Proposed Methodology:

Bot Detection Module

The bot detection module models Twitter as a directed graph, where users are represented as nodes and follower–following relationships as edges. Each user is associated with a feature vector constructed from metadata, behavioral statistics, and aggregated tweet information.

Feature engineering includes user-level attributes (e.g., followers, following, tweet count), derived ratios (e.g., follower–following ratio, engagement rates), and content-based aggregates (e.g., hashtag usage, mentions, URLs). Graph-based features such as node degree are also incorporated. All features are normalized before training.

A three-layer Graph Neural Network based on GraphSAGE is employed to learn node representations by aggregating neighborhood information. The model uses batch normalization, dropout, and residual connections to improve generalization. Training is performed using a weighted cross-entropy loss with label smoothing to address class imbalance.

Polarization Module

The polarization module analyzes ideological patterns using semantic representations of text. Each text sample is encoded into a dense vector using a pre-trained Sentence Transformer model (all-MiniLM-L6-v2), capturing contextual semantic information.

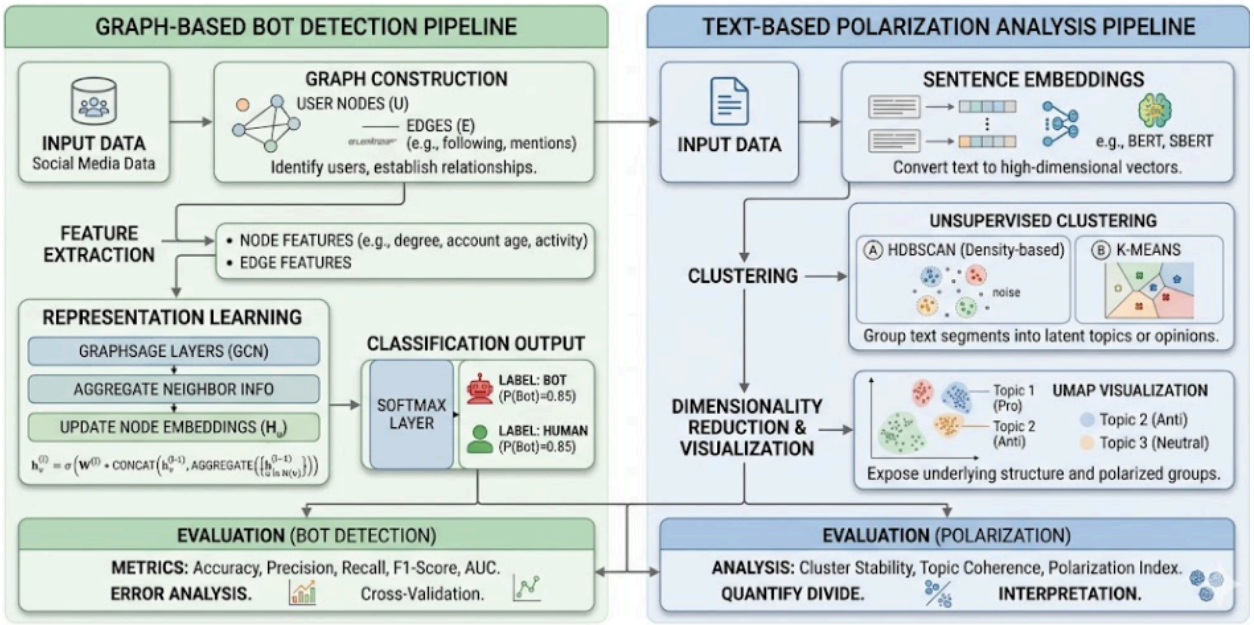
Clustering is performed using two approaches: HDBSCAN, which identifies clusters of varying density and handles noise, and KMeans, used as a baseline method with a fixed number of clusters. To facilitate visualization, embeddings are projected into a lower-dimensional space using UMAP.

Polarization is quantified using inter-cluster and intra-cluster distances. A polarization index and an echo chamber strength metric are computed to measure ideological separation and cluster cohesion. Additionally, intervention experiments are conducted by applying random removal and structured removal of dominant clusters to analyze changes in polarization.

Architecture:

System Overview

The overall system consists of two independent pipelines: a graph-based pipeline for bot detection and a semantic pipeline for polarization analysis. Both modules operate on different data modalities but follow a structured processing workflow.



Dataset:

Bot Detection Dataset:

The bot detection module utilizes the **TwiBot-22 benchmark dataset**, which models Twitter as a graph with multiple entity types. The dataset includes user information (user.json), tweet data (tweet_0.json–tweet_8.json), relational edges (edge.csv), and ground-truth labels (label.csv).

In this study, a user-centric graph is constructed using follower–following relationships from edge.csv. Users are labeled as human or bot, and a balanced subset is sampled for training. Tweet data is aggregated at the user level to extract behavioral features. The final graph consists of user nodes, directed edges, and binary labels.

Polarization Dataset:

The polarization analysis module utilizes the **USC X 2024 US Election Twitter/X Dataset,** which contains large-scale tweet data related to the 2024 U.S. elections. The dataset is organized into multiple partitions, each consisting of chunked files with approximately 50,000 tweets per file. Each entry includes text and associated metadata; however, this study uses only the textual content for semantic analysis.

A labeled subset with ideological annotations (*Left* or *Right*) is used, provided in training, validation, and test splits, which are combined into a single corpus for unsupervised clustering. Ideology labels are encoded as binary values (Left = 0, Right = 1) and are used only for evaluation purposes, such as clustering alignment and purity metrics.

Experimental Setup:

All experiments were conducted using Python with PyTorch and PyTorch Geometric. The bot detection model is based on a three-layer GraphSAGE architecture with 256 hidden units, trained using the AdamW optimizer (learning rate = 0.002, weight decay = 1×10^{-4}). The model was trained for up to 200 epochs with early stopping based on validation performance.

For polarization analysis, text embeddings were generated using the all-MiniLM-L6-v2 model. Clustering was performed using HDBSCAN (minimum cluster size = 10) and KMeans (k = 5). UMAP was used for dimensionality reduction with cosine distance.

Model evaluation was performed on a held-out test set using standard classification and clustering metrics.

Results:

Bot Detection Module:

The GraphSAGE model achieves an overall accuracy of 0.76 and a ROC-AUC score of 0.8524, indicating strong discriminative performance. As shown in Table 1, the model attains higher recall for the bot class (0.83), suggesting effective identification of automated accounts. However, the comparatively lower precision (0.69) indicates the presence of some false positives. This trade-off reflects the model's emphasis on capturing bot behavior, which is critical in detection tasks.

Table 1: Bot Detection Performance

Class	Precision	Recall	F1-Score	Support
Human (0)	0.83	0.70	0.76	529
Bot (1)	0.69	0.83	0.75	431

Table 2: Overall Performance

Metric	Value
Accuracy	0.76
Macro Avg F1	0.76
Weighted Avg F1	0.76
ROC-AUC	0.8524

Polarization Module:

The clustering results demonstrate that HDBSCAN outperforms KMeans in terms of cluster separation, as reflected by a higher silhouette score (0.0678 vs 0.0329). However, the low Adjusted Rand Index (0.0037) indicates weak alignment with ground-truth ideology, suggesting that ideological patterns are not strictly separable in the embedding space. Despite this, a cluster purity of 0.7155 indicates that clusters still exhibit a degree of ideological consistency.

Table 3: Clustering Metrics:

Metric	HDBSCAN	KMeans
Silhouette Score	0.0678	0.0329
ARI	0.0037	—
Cluster Purity	0.7155	—

The computed polarization index (0.3361) indicates moderate separation between ideological groups. The relatively higher intra-cluster distance (0.7560) compared to inter-cluster distance (0.3826) suggests that clusters are internally diverse. The echo chamber strength (0.5062) further reflects limited but noticeable clustering of similar viewpoints.

Table 4: Polarization Analysis

Metric	Value
Polarization Index	0.3361
Inter-cluster Distance	0.3826
Intra-cluster Distance	0.7560
Echo Chamber Strength	0.5062

The intervention experiments reveal distinct effects on polarization. Random removal increases the polarization index to 0.4660, indicating that removing noise can enhance separation between clusters. In contrast, structured removal reduces polarization to zero, demonstrating that dominant clusters play a critical role in maintaining ideological structure.

Table 5: Effect of Interventions on Polarization

Scenario	Polarization
Baseline	0.3361
Random Removal	0.4660
Structured Removal	0.0000

Limitations:

- The bot detection module uses a static graph representation and does not capture the temporal dynamics of user behavior.
- The polarization module relies only on semantic embeddings and does not incorporate user interaction networks.
- The framework captures semantic echo chambers but not network-based echo chambers.
- The two modules operate independently and are not jointly optimized.

Future Work:

- Integrate graph-based and text-based representations into a unified model.
- Incorporate temporal dynamics to analyze the evolution of behavior and polarization.
- Develop hybrid architectures (e.g., Graph Attention Networks or transformer-based graph models).
- Improve interpretability using attention mechanisms or feature attribution methods.

Conclusion:

This study presents a dual-module framework for analyzing social media dynamics through graph-based bot detection and semantic polarization analysis. The GraphSAGE model demonstrates effective performance in identifying automated accounts by leveraging structural and behavioral features. The polarization module reveals meaningful clustering patterns in semantic space and introduces quantitative metrics to measure ideological separation, indicating the presence of semantic echo chambers.

Intervention experiments highlight the role of dominant clusters in shaping polarization, showing that targeted removal strategies can significantly alter ideological structure. Although the modules operate independently, the framework provides a structured approach for analyzing both user behavior and content semantics, contributing to a broader understanding of social media ecosystems.

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